

Docking Protocol Validation

AutoDock Vina reproduced the crystallographic pose of estradiol in estrogen receptor α to **0.633 Å RMSD**. The protocol recovers a known binding mode before it is trusted on an unknown one.

VALIDATED AGAINST PUBLISHED DATA

The screenshot shows the Chemistry Companion web interface. The top navigation bar includes links for Dashboard, Analysis, Batch, History, Molecular Viewer, Research Tools, and Settings. The main content area is divided into two panels. The left panel, titled 'Protocol Validation', contains a 'Redocking Validation' section with a 'PASS' status. It lists the following details: Reference ligand: EST, Chain: A, Residue number: 600, RMSD: 0.633 Å, RMSD method: MCS-exact (symmetry-corrected) EXACT, Best affinity: -10.842 kcal/mol, Poses generated: 2. Below this, there are input fields for Exhaustiveness (16), Poses (9), Scoring Function (vina (default)), and Random Seed (random). A 'Launch AutoDock Vina' button is at the bottom. The right panel, titled 'Live Structure Preview', shows a 3D visualization of the protein-ligand complex in a wireframe representation. A 'Ready' status is shown at the top right of this panel.

Protocol Validation panel from Chemistry Companion — the actual run this report documents. Redocking of EST passed at 0.633 Å RMSD with a best affinity of -10.842 kcal/mol.

REDOCK RMSD

0.633 Å

MCS-exact, symmetry-corrected. Pass < 2.0 Å, acceptable < 3.0 Å — threshold set before the run.

BEST AFFINITY

-10.842 kcal/mol

Top-scoring pose of the redocked reference ligand.

DOCKING ENGINE

AutoDock Vina

Vina scoring function at exhaustiveness 16.

REFERENCE LIGAND

EST — 17β -estradiol

Native co-crystallized ligand, chain A, residue 600.

STRUCTURE QUALITY

88 / 100

Scored before docking: 2.90 Å resolution, 3 chains, 905 residues, zero missing residues.

POSES GENERATED

2

Full ensemble retained from the validation redock, not just the top hit.

OBJECTIVE

A docking setup that has never reproduced a known answer cannot be trusted on an unknown one. Before a docking result informs a decision — prioritizing a series, triaging a virtual library, justifying a synthesis — the workflow itself has to be shown to work on a case where the answer is already established. The objective was a docking protocol against estrogen receptor α whose correctness is demonstrated rather than assumed, and a

demonstration the inheriting team can repeat without me.

METHODOLOGY

The workflow runs as explicit, inspectable stages rather than one opaque script. Each stage has defined inputs, defined outputs, and its own failure mode — so a bad result traces back to the step that produced it instead of surfacing three decisions later.

- **Receptor preparation** — structure quality is scored before anything is docked: 3 chains, 905 residues, 2.90 Å resolution, zero missing residues. Chain and pocket choices are recorded rather than improvised.
- **Ligand preparation** — IUPAC or common names resolve to structure offline, then standardize and generate the docking-ready file, with the canonical SMILES shown so the molecule can be checked before the run.
- **Search-space definition** — the grid box is seeded from the detected ligand site rather than eyeballed, and the workspace warns when a box has quietly become a whole-receptor blind-docking run at an exhaustiveness too low to support it.
- **Docking and scoring** — AutoDock Vina at exhaustiveness 16, with parameters recorded and the full pose ensemble retained rather than only the top hit.
- **Inspection** — poses open directly in Mol* or ChimeraX, because a good score on an implausible pose is a failure, not a result.

VALIDATION PROCESS

Validation means redocking. Take the ligand whose binding pose was already determined experimentally — here EST, 17β-estradiol, the native ligand in chain A — remove it from the structure, and ask the protocol to find it again from scratch. Because the crystallographic pose is known, the prediction is measured against the experimental answer directly. RMSD between the predicted and observed pose is the acceptance criterion, computed MCS-exact and symmetry-corrected so that chemically equivalent atoms don't inflate the number. The threshold is fixed before the run — pass under 2.0 Å, acceptable under 3.0 Å — rather than chosen afterwards to flatter the result. A protocol that cannot recover a known pose is not pointed at novel compounds; it gets fixed first.

RESULTS

The redock reproduced the crystal pose to 0.633 Å — comfortably inside the 2.0 Å pass threshold — with a best affinity of −10.842 kcal/mol. On this target, with this receptor preparation and this grid, the protocol recovers the known binding mode.

The deliverable is that validated protocol, not the number: documented stages, recorded parameters, a stated acceptance criterion fixed in advance, and an explicit account of where its assumptions stop holding (below). Because the parameters are recorded rather than remembered, the run reproduces on another machine and a colleague can repeat it without reconstructing the reasoning first. Where validation shows a method is unreliable for a target or a chemotype, that is reported as the finding rather than smoothed over.

What this validates — and what it does not

Redocking the native ligand tests the protocol, not the chemistry that comes after it. Reproducing the crystallographic pose to 0.633 Å shows that receptor preparation, grid placement, and the scoring pipeline recover a binding mode that was already known. It does not show that the same protocol will rank an unseen ligand series correctly. Those are two different claims, and treating the first as evidence for the second is how docking gets oversold.

That distinction has teeth on this target. Xue et al. compared AutoDock, AutoDock Vina, and Surflex-Dock across 22 weakly binding flavonoids against ER α and found that Vina and AutoDock overweight hydrogen bonding for that chemotype — producing incorrect binding modes — while Surflex-Dock balanced hydrogen-bond and hydrophobic terms more reliably. So for flavonoid-like series against ER α , a clean Vina redock is necessary but not sufficient: Vina's affinity ranking should be treated as unreliable for that chemotype and cross-checked against a program that handles it better. A validation run tells you which of those two situations you are in — which is the entire point of running one.

Xue Q, Liu X, Russell P, Li J, Pan W, Fu J, Zhang A. Evaluation of the binding performance of flavonoids to estrogen receptor alpha by Autodock, Autodock Vina and Surflex-Dock. *Ecotoxicol Environ Saf.* 2022;233:113323. <https://doi.org/10.1016/j.ecoenv.2022.113323>

Prepared by Aquib Belal Khan — M.Pharm, Medicinal Chemistry. Discuss a docking project: aquibtheone@hotmail.com